Assignment Guidelines

EN.705.601 Applied Machine Learning

Overview

The course goal is introducing machine learning pipeline design and analysis as a branch of study in computer science, including developing the necessary mathematical and computational skills, surveying the major design paradigms, and analyzing several specific algorithms and libraries. Therefore, a hands-on experience is clearly required and assignments are the best vehicle in this goal.

Every module will contain Assignments with problems sets that you will complete individually. Few questions will be completed by giving direct answers and most questions will be completed by a machine learning pipeline using a Jupyter notebook using your own code and certain open source libraries such as `pandas`, `numpy`, `scikit-learn`, etc. Generally, at the end of each assignment, there is one or two conclusion/evaluation question(s) about the problem at hand and about your pipeline's performance. A maximum of 100 points per problem set can be earned, with partial credit given as discussed below. Each problem set will be **due by Day 7** of the module in which they are assigned. **No late problems sets will be accepted without prior approval.**

Quizzes encourage the student to watch the module videos and follow the material in their studying and learning activities. There might be questions which may look irrelevant but these have to be taken with the intent of measuring the amount of attention student pays to the course materials. Students who watch the videos should be able to achieve full score easily.

Submission

The notebook has the code and results, such as plots to show work, justify your answers.

Answer assignment questions (such as, "which classifier has the highest performance and why?") with text written in **Markdown cells**. (Do not use Code section or Python comments. These comments help you, the reader to check/remind your programming work)

Submit each assignment as a Jupyter Notebook in original format (extension **`ipynb`**) file to the corresponding Assignment link within each module.

Use the following for naming your submission **`Last Name-AssignX.ipynb`**, where **`X`** is the number of the module, i.e. `1, 2, ..., 12, 13, 14`.

Important: Before submitting your notebook assignments, please use **Kernel → Restart & Run All**. This will result running the notebook from reset. Then save and submit that copy. (Otherwise, some Code cells may not function when opened by your peers or the grader.)

If the notebook was not run from Restart as described earlier, the grader or Instructor may reject your submission.



Assignment Grading Considerations

Each Module Assignment will be worth a total of 100 points, divided among the problems in each set depending on their length and/or difficulty, and indicated at the beginning of the question. Problems will be graded for relevancy, accuracy, and completeness. The submitted Jupyter notebook is expected to run fully from the very beginning to the end after a clean kernel is started. Markdown cells will be used to answer questions, they can be used to explain the approach, and Code comments must be included in the pipeline to present the audience or grader about the pipeline. The amount of explanations and comments must be reasonable like any other programming assignment. Solutions without justification will be not be considered complete and the grade will be adjusted down accordingly.

A note on academic dishonesty: While discussion of the homework is allowed, the assignments are intended to be done individually and you must prepare your solutions separately. If a code excerpt is used from other resources, then that code has to be paraphrased or made "your own" similar to writing a research paper. Direct copying of written work or computer code is considered to be dishonest and will result in a grade of zero on the assignment and possible an F in the course.

Sources: References are preferred to be actual publications or "quality" web sites. Stack overflow, Kaggle, university web sites and similar technical web sites with no commercial advertisements are considered acceptable.

Instructor Role

The Assignment portion of the total course grade is the largest single contribution because it permits students to demonstrate their mastery of the concepts and their ability to program demonstrations. Therefore, I will be looking for consistent and dedicated attention to the problems every week. I will accept and answer questions (by Teams – preferred, by email, during Office Hours, or occasionally on the telephone) on the approach to any of the problems. In general, any responses to questions about the problems will be shared with the class so that everyone has the same information. Problem sets will be graded and returned weekly so that students may monitor their progress throughout the course. Detailed feedback will be provided on each assignment so that the student clearly understands any errors made and how to correct them.